UNIVERSIDAD TECNOLÍGICA

TULA-TEPEJI

Tecnologías de la Información Desarrollo De Software Multiplataforma

Ejercicios SQL

Desarrollado por:

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Lección 1.

properties of each movie. To continue onto the next lesson, alter the query to find the exact information we need for each task.

Table: Movies

Id	Title	Director	Year	Length_minutes
1	Toy Story	John Lasseter	1995	81
2	A Bug's Life	John Lasseter	1998	95
3	Toy Story 2	John Lasseter	1999	93
4	Monsters, Inc.	Pete Docter	2001	92
5	Finding Nemo	Andrew Stanton	2003	107
6	The Incredibles	Brad Bird	2004	116
7	Cars	John Lasseter	2006	117
8	Ratatouille	Brad Bird	2007	115
9	WALL-E	Andrew Stanton	2008	104
10	Up	Pete Docter	2009	101

```
SELECT * FROM movies;
/*Elaborado por Adair Jesús Cruz Huitrón*/
```

RESET

Exercise 1 — Tasks

1. Find the `title` of each film ✓
2. Find the `director` of each film ✓
3. Find the `title` and `director` of each film ✓
4. Find the `title` and `year` of each film ✓
5. Find `all` the information about each film ✓

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Next – SQL Lesson 2: Queries with constraints (Pt. 1)
Previous – Introduction to SQL

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Lección 2.

Exercise

Using the right constraints, find the information we need from the **Movies** table for each task below.

Table: Movies

Id	Title	Director	Year	Length_minutes
1	Toy Story	John Lasseter	1995	81
2	A Bug's Life	John Lasseter	1998	95
3	Toy Story 2	John Lasseter	1999	93
4	Monsters, Inc.	Pete Docter	2001	92
5	Finding Nemo	Andrew Stanton	2003	107

```
SELECT *
FROM movies
WHERE id<6;
/* Elaborado por Adair Jesús Cruz Huitrón */
```

RESET

Exercise 2 — Tasks

1. Find the movie with a row `id` of 6 ✓
2. Find the movies released in the `year`s between 2000 and 2010 ✓
3. Find the movies **not** released in the `year`s between 2000 and 2010 ✓
4. Find the first 5 Pixar movies and their release `year` ✓

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Lección 3.

```
WHERE condition
    AND/OR another_condition
    AND/OR …;
```

Table: Movies

Id	Title	Director	Year	Length_minutes
9	WALL-E	Andrew Stanton	2008	104
87	WALL-G	Brenda Chapman	2042	97

Exercise 3 — Tasks

1. Find all the Toy Story movies ✓
2. Find all the movies directed by John Lasseter ✓
3. Find all the movies (and director) not directed by John Lasseter ✓
4. Find all the WALL-* movies ✓

```
SELECT *
FROM movies
WHERE Title LIKE "WALL-_";
/* Desarrollado por Adair Jesús Cruz Huitrón */
```

RESET

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Lección 4.

There are a few concepts in this lesson, but all are pretty straight-forward to apply. To spice things up, we've gone and scrambled the **Movies** table for you in the exercise to better mimic what kind of data you might see in real life. Try and use the necessary keywords and clauses introduced above in your queries.

Table: Movies

Id	Title	Director	Year	Length_minutes
7	Monsters University	Dan Scanlon	2013	110
14	Monsters, Inc.	Pete Docter	2001	92
2	Ratatouille	Brad Bird	2007	115
12	The Incredibles	Brad Bird	2004	116
13	Toy Story	John Lasseter	1995	81

Exercise 4 — Tasks

1. List all directors of Pixar movies (alphabetically), without duplicates ✓
2. List the last four Pixar movies released (ordered from most recent to least) ✓
3. List the **first** five Pixar movies sorted alphabetically ✓
4. List the **next** five Pixar movies sorted alphabetically ✓

```
SELECT *
FROM movies
ORDER BY title ASC
LIMIT 5 OFFSET 5;
/* Desarrollado por Adair Jesús Cruz Huitrón */
```

RESET

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Lección 5.

Try and write some queries to find the information requested in the tasks below. You may have to use a different combination of clauses in your query for each task. Once you're done, continue onto the next lesson to learn about queries that span multiple tables.

Table: North_american_cities

City	Country	Population	Latitude	Longitude
Chicago	United States	2718782	41.878114	-87.629798
Houston	United States	2195914	29.760427	-95.369803

Review 1 — Tasks

1. List all the Canadian cities and their populations ✓
2. Order all the cities in the United States by their latitude from north to south ✓
3. List all the cities west of Chicago, ordered from west to east ✓
4. List the two largest cities in Mexico (by population) ✓
5. List the third and fourth largest cities (by population) in the United States and their population ✓

```
SELECT *
FROM north_american_cities
WHERE COUNTRY LIKE "United States"
ORDER BY population DESC
Limit 2 OFFSET 2;
/* Desarrollado por Adair Jesús Cruz Huitrón */
```

RESET

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Lección 6.

4	Monsters, Inc.	Pete Docter	2001	92	12	6.4	191452396	368400000
5	Finding Nemo	Andrew Stanton	2003	107	3	7.9	245852179	239163000
6	The Incredibles	Brad Bird	2004	116	6	8	261441092	370001000

Query Results

Id	Title	Director	Year	Length_minutes	Movie_id	Rating	Domestic_sales	International_sal
9	WALL-E	Andrew Stanton	2008	104	9	8.5	223808164	297503696
11	Toy Story 3	Lee Unkrich	2010	103	11	8.4	415004880	648167031
1	Toy Story	John Lasseter	1995	81	1	8.3	191796233	170162503
10	Up	Pete Docter	2009	101	10	8.3	293004164	438338580
5	Finding Nemo	Andrew Stanton	2003	107	5	8.2	380843261	555900000
4	Monsters, Inc	Pete Docter	2001	92	4	8.1	289916256	272900000

Exercise 6 — Tasks

1. Find the domestic and international sales for each movie ✓
2. Show the sales numbers for each movie that did better internationally rather than domestically ✓
3. List all the movies by their ratings in descending order ✓

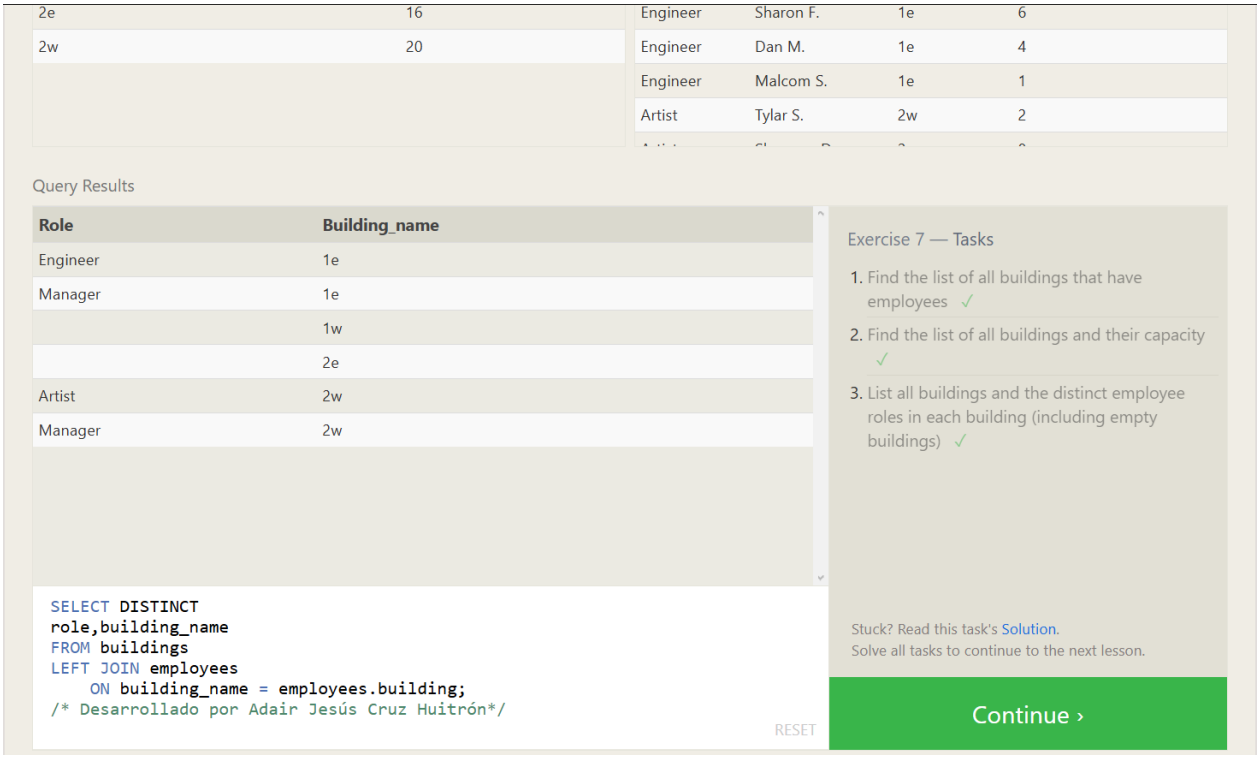
```
SELECT *, boxoffice.rating
FROM movies
INNER JOIN boxoffice
    ON movies.id = boxoffice.movie_id
ORDER BY rating DESC;
/* Desarrollado por Adair Jesús Cruz Huitrón */
```

RESET

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Lección 7.



2e	16	Engineer	Sharon F.	1e	6
2w	20	Engineer	Dan M.	1e	4
		Engineer	Malcom S.	1e	1
		Artist	Tylar S.	2w	2

Query Results

Role	Building_name
Engineer	1e
Manager	1e
	1w
	2e
Artist	2w
Manager	2w

```
SELECT DISTINCT
role,building_name
FROM buildings
LEFT JOIN employees
    ON building_name = employees.building;
/* Desarrollado por Adair Jesús Cruz Huitrón*/
```

RESET

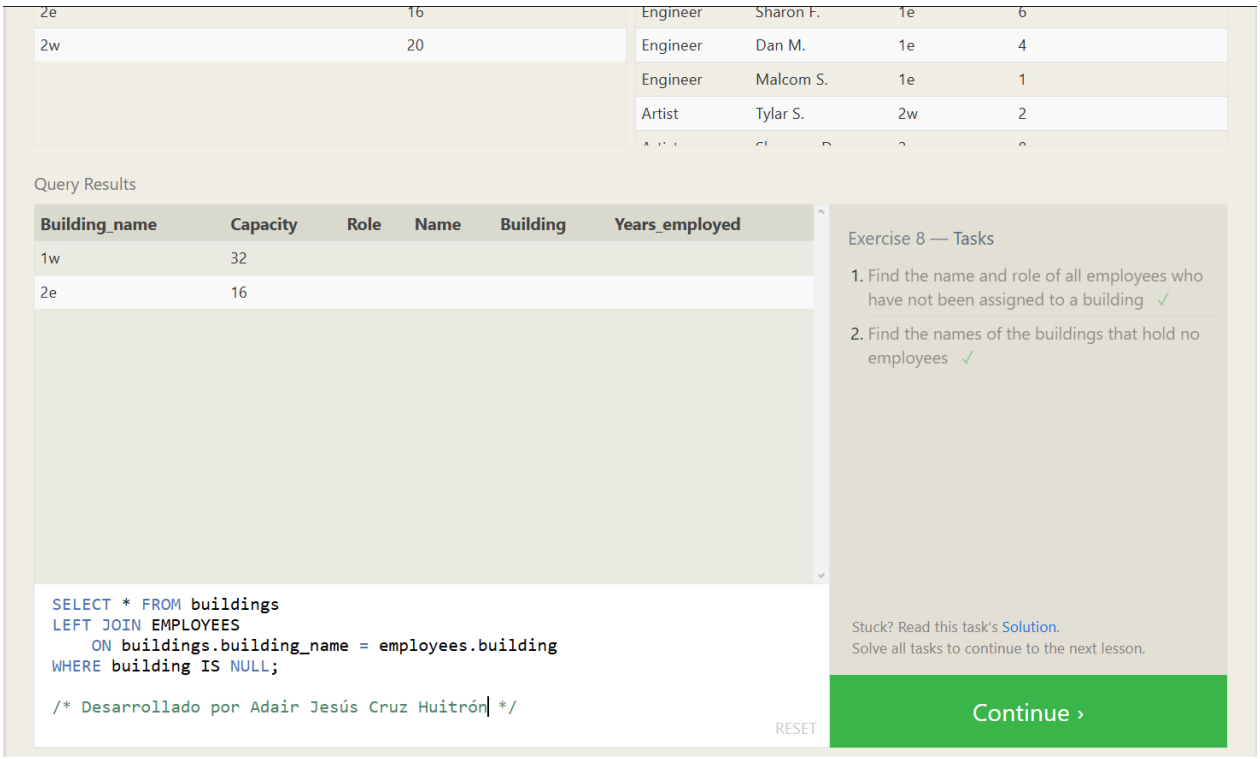
Exercise 7 — Tasks

1. Find the list of all buildings that have employees ✓
2. Find the list of all buildings and their capacity ✓
3. List all buildings and the distinct employee roles in each building (including empty buildings) ✓

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Lección 8.



2e	16	Engineer	Sharon F.	1e	6
2w	20	Engineer	Dan M.	1e	4
		Engineer	Malcom S.	1e	1
		Artist	Tylar S.	2w	2

Query Results

Building_name	Capacity	Role	Name	Building	Years_employed
1w	32				
2e	16				

```
SELECT * FROM buildings
LEFT JOIN EMPLOYEES
    ON buildings.building_name = employees.building
WHERE building IS NULL;

/* Desarrollado por Adair Jesús Cruz Huitrón */
```

RESET

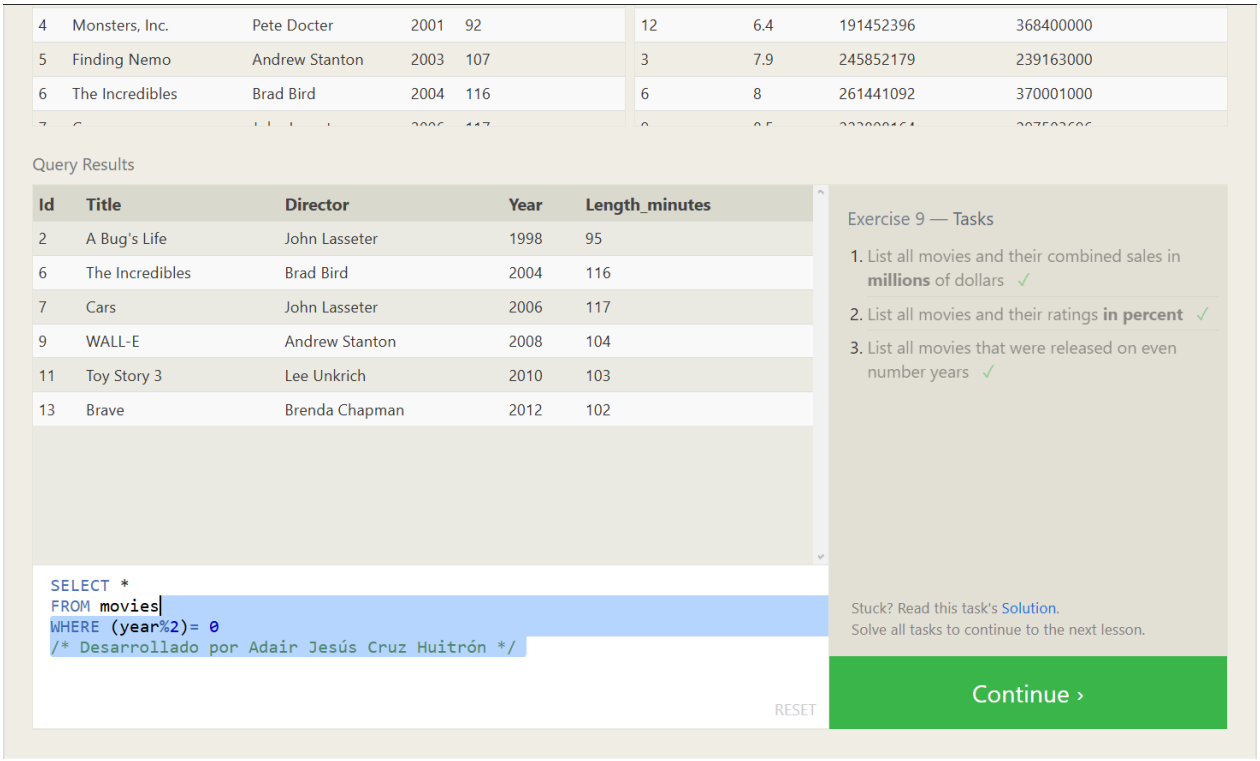
Exercise 8 — Tasks

1. Find the name and role of all employees who have not been assigned to a building ✓
2. Find the names of the buildings that hold no employees ✓

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Lección 9.



4	Monsters, Inc.	Pete Docter	2001	92	12	6.4	191452396	368400000
5	Finding Nemo	Andrew Stanton	2003	107	3	7.9	245852179	239163000
6	The Incredibles	Brad Bird	2004	116	6	8	261441092	370001000

Query Results

Id	Title	Director	Year	Length_minutes
2	A Bug's Life	John Lasseter	1998	95
6	The Incredibles	Brad Bird	2004	116
7	Cars	John Lasseter	2006	117
9	WALL-E	Andrew Stanton	2008	104
11	Toy Story 3	Lee Unkrich	2010	103
13	Brave	Brenda Chapman	2012	102

Exercise 9 — Tasks

1. List all movies and their combined sales in **millions** of dollars ✓
2. List all movies and their ratings **in percent** ✓
3. List all movies that were released on even number years ✓

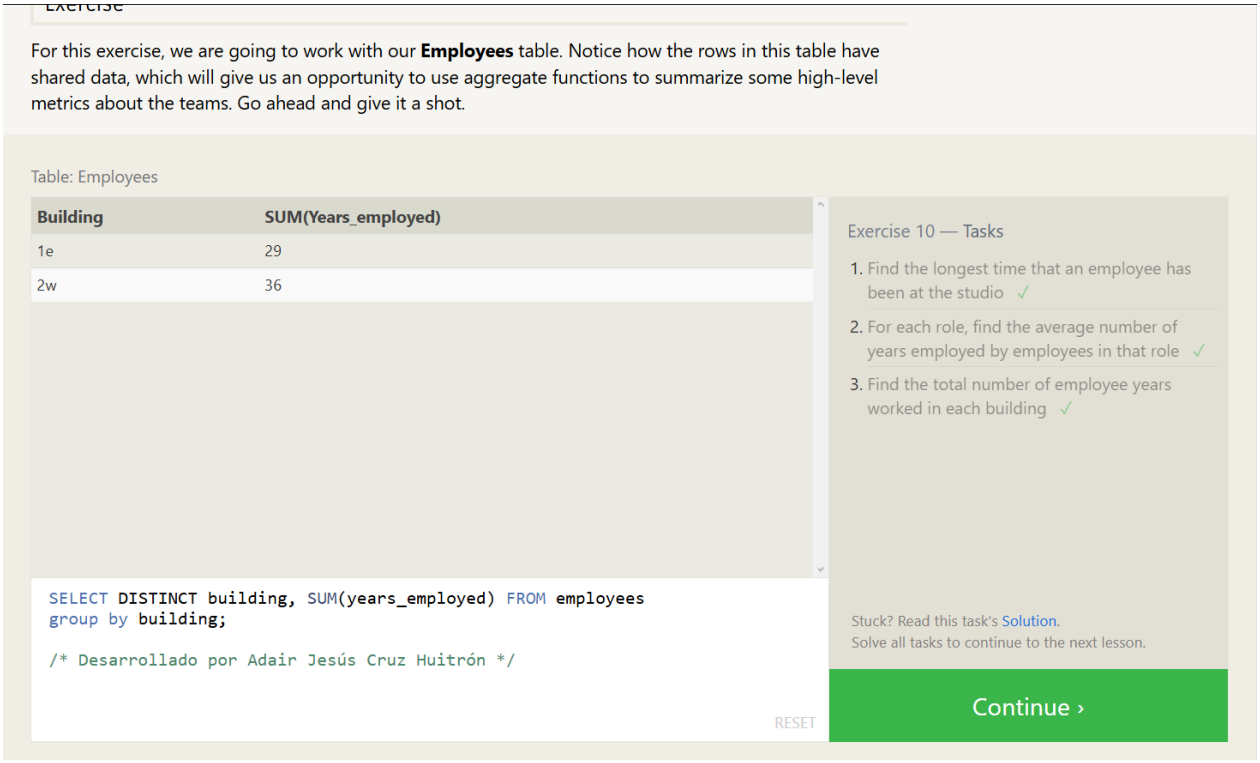
```
SELECT *
FROM movies
WHERE (year%2)= 0
/* Desarrollado por Adair Jesús Cruz Huitrón */
```

RESET

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Lección 10.



For this exercise, we are going to work with our **Employees** table. Notice how the rows in this table have shared data, which will give us an opportunity to use aggregate functions to summarize some high-level metrics about the teams. Go ahead and give it a shot.

Table: Employees

Building	SUM(Years_employed)
1e	29
2w	36

Exercise 10 — Tasks

1. Find the longest time that an employee has been at the studio ✓
2. For each role, find the average number of years employed by employees in that role ✓
3. Find the total number of employee years worked in each building ✓

```
SELECT DISTINCT building, SUM(years_employed) FROM employees
group by building;

/* Desarrollado por Adair Jesús Cruz Huitrón */
```

RESET

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Lección 11.

Exercise

For this exercise, you are going to dive deeper into **Employee** data at the film studio. Think about the different clauses you want to apply for each task.

Table: Employees

Role	Years Employeed
Engineer	17

```
SELECT role, sum(years_employed) AS [years employeed] FROM employees
GROUP BY role
HAVING role = "Engineer";
/* Desarrollado por Adair Jesús Cruz Huitrón*/
```

RESET

Exercise 11 — Tasks

1. Find the number of Artists in the studio (without a **HAVING** clause) ✓
2. Find the number of Employees of each role in the studio ✓
3. Find the total number of years employed by all Engineers ✓

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Lección 12.

4	Monsters, Inc.	Pete Docter	2001	92	12	6.4	191452396	368400000
5	Finding Nemo	Andrew Stanton	2003	107	3	7.9	245852179	239163000
6	The Incredibles	Brad Bird	2004	116	6	8	261441092	370001000

Query Results

Director	Movies	Sales
Andrew Stanton	2	1458055121
Brad Bird	2	1255164910
Brenda Chapman	1	538983207
Dan Scanlon	1	743559607
John Lasseter	5	2232208025
Lee Unkrich	1	1063171911
Pete Docter	2	1294159000

```
SELECT director,count(*) as [movies], (SUM(domestic_sales)+SUM
    (international_sales)) AS [sales] FROM movies
INNER JOIN boxoffice
    ON id = movie_id
GROUP BY director;
/* Desarrollado por Adair Jesús Cruz Huitrón */
```

RESET

Exercise 12 — Tasks

1. Find the number of movies each director has directed ✓
2. Find the total domestic and international sales that can be attributed to each director ✓

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Lección 13.

15	Toy Story 4	John Lasseter	2026	214	15	8.7	340000000	270000000

Query Results

Movie_id	Rating	Domestic_sales	International_sales
3	7.9	245852179	239163000
1	8.3	191796233	170162503
2	7.2	162798565	200600000
15	8.7	340000000	270000000

```
INSERT INTO boxoffice
VALUES
(15, 8.7, (340*1000000),(270*1000000));
/* Desarrollado por Adair Jesús Cruz Huitrón */
```

RUN QUERY RESET

Exercise 13 — Tasks

1. Add the studio's new production, **Toy Story 4** to the list of movies (you can use any director) ✓
2. Toy Story 4 has been released to critical acclaim! It had a rating of **8.7**, and made **340 million domestically** and **270 million internationally**. Add the record to the `BoxOffice` table. ✓

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Lección 14.

Exercise

It looks like some of the information in our **Movies** database might be incorrect, so go ahead and fix them through the exercises below.

Table: Movies

4	Monsters, Inc.	Pete Docter	2001	92
5	Finding Nemo	Andrew Stanton	2003	107
6	The Incredibles	Brad Bird	2004	116
7	Cars	John Lasseter	2006	117
8	Ratatouille	Brad Bird	2007	115
9	WALL-E	Andrew Stanton	2008	104
10	Up	Pete Docter	2009	101
11	Toy Story 3	Lee Unkrich	2010	103
12	Cars 2	John Lasseter	2011	120
13	Brave	Brenda Chapman	2012	102
14	Monsters University	Dan Scanlon	2013	110

```
UPDATE movies
SET
    Title = "Toy Story 3",
    Director = "Lee Unkrich"
WHERE id = 11
/* Desarrollado por Adair Jesús Cruz Huitrón */
```

RUN QUERY RESET

Exercise 14 — Tasks

1. The director for A Bug's Life is incorrect, it was actually directed by **John Lasseter** ✓
2. The year that Toy Story 2 was released is incorrect, it was actually released in **1999** ✓
3. Both the title and director for Toy Story 8 is incorrect! The title should be "Toy Story 3" and it was directed by **Lee Unkrich** ✓

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›

Lección 15.

Exercise

The database needs to be cleaned up a little bit, so try and delete a few rows in the tasks below.

Table: Movies

Id	Title	Director	Year	Length_minutes
7	Cars	John Lasseter	2006	117
8	Ratatouille	Brad Bird	2007	115
10	Up	Pete Docter	2009	101
11	Toy Story 3	Lee Unkrich	2010	103
12	Cars 2	John Lasseter	2011	120
13	Brave	Brenda Chapman	2012	102
14	Monsters University	Dan Scanlon	2013	110

```
DELETE FROM movies;
WHERE Director = "Andrew Stanton"
/* Desarrollado por Adair Jesús Cruz Huitrón */
```

RUN QUERY RESET

Exercise 15 — Tasks

1. This database is getting too big, lets remove all movies that were released **before** 2005. ✓
2. Andrew Stanton has also left the studio, so please remove all movies directed by him. ✓

Stuck? Read this task's Solution.
Solve all tasks to continue to the next lesson.

Continue ›